Ladders and ropes

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The eighteen participants of the Animath camp each stand at the foot of a ladder, at the top of which lies an envelope containing a precious exercise. The organisers have stretched finitely many ropes between some of the ladders. Assume that two ropes are never attached to the same rung. The trainees start climbing their ladders and systematically take every rope they meet, whether it goes up or down.

Show that all the exercises will be solved (since it is trivially a formality for the pupils to get to the bottom of these subtle statements).